

Research



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Global change biology

The past, present and future of cleaner fish cognitive performance as a function of CO₂ levels

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Ocean acidification is one of the many consequences of climate change. Various studies suggest that marine organisms' behaviour will be impaired under high CO₂. Here, we show that the cognitive performance of the cleaner wrasse, *Labroides dimidiatus*, has not suffered from the increase of CO₂ from pre-industrial levels to today, and that the standing variation in CO₂ tolerance offers potential for adaptation to at least 750 µatm. We acclimated cleaners over 30 days to five levels of pCO₂, from pre-industrial to high future CO₂ scenarios, before testing them in an ecologically relevant task—the ability to learn to prioritize an ephemeral food source over a permanent one. Fish learning abilities remained stable from pre-industrial to present-day pCO₂. While performance was reduced under mid (750 µatm) and high CO₂ (980 µatm) scenarios, under the former 36% of cleaners still solved the task. The presence of tolerant individuals reveals potential for adaptation, as long as selection pressure on cognitive performance is strong. However, the apparent absence of high CO₂ tolerant fish, and potentially synergistic effects between various climate change stressors, renders the probability of further adaptation unlikely.

1. Introduction

Climate change-related stressors, such as warming, acidification and intensification of extreme weather, have serious negative impacts on marine ecosystems [1,2]. Ocean acidification is a key climate change-induced stressor to coral reefs as it affects calcification [3]. Yet, acidification impacts are more widespread along multiple marine taxa. In particular, acidification can impair fish behaviour and learning by malfunction of different sensory mechanisms, such as vision, hearing and olfaction [4–8]. The putative mechanism for this behavioural disruption is CO₂ interference on GABAergic neurotransmission, where changes in [Cl⁻] and [HCO₃⁻] in plasma and neurons, as result of acid–base regulation, cause an alteration in GABA_A receptor function [9–11]. Nonetheless, species are known to be able to adapt to environmental change, either through selection of standing genetic variation or via mutations [12]. So far, some evolutionary studies indicated that adaptation to partly compensate for the adverse effects of ocean acidification is evident in planktonic life after a few hundred generations [13,14]. Therefore, regarding fish behavioural disruptions in high CO₂, the question is not whether contemporary fish malfunction under future CO₂ concentrations but whether adaptation can occur rapidly enough to cope with the pace of increasing acidification.

Multiple studies have shown that transgenerational exposure to ocean acidification is not enough to diminish all its effects on fish behaviour, but variation in tolerance among individuals could be important for putative adaptive responses [15,16]. Unfortunately, many fish species are rather unsuitable subjects for selection experiments due to long generation times and/or complex reproductive cycles making it virtually impossible to complete their life cycle under laboratory conditions [13]. Thus, alternative testing methods need to be established to include the evolutionary aspect in climate change scenarios for these important clades.

When trying to incorporate natural selection into predictions of future scenarios of fish behaviour, it is important to realize that today's conditions are already different from pre-industrial times [12]. pCO₂ levels in the oceans increased by about 46.9% from 275 µatm to present-day 404 µatm [17]. Therefore, to control for already occurring impairments under present-day conditions, ocean acidification experiments should include a treatment of pre-industrial levels. If not, the researchers are assuming that the studied species was apparently able to cope with the changes so far. Second, the data on performance under future scenarios should not focus on mean performance but on individual performance as directional selection operates on the extremes and not on the mean [13,18,19].

Here, we evaluate the adaptive potential to ocean acidification in a cognitive experiment on cleaner fish *Labroides dimidiatus* (hereafter 'cleaner'). Cleaners remove ectoparasites, dead skin and mucus from 'client' fishes [20]. A major conflict of interest arises because cleaners prefer eating the protective mucus from their clients, which constitutes cheating [21]. This is the likely cause for the highly sophisticated decision-rules used by cleaners during interactions, including reconciliation, reputation management and social tool-use [21,22]. In addition, cleaners are highly sensitive to the partner choice options of their clients. As cleaners have about 2000 interactions per day [23,24], simultaneous invitations for inspection by two or more clients are frequent. In such cases, cleaners prioritize visitor clients with access to alternative cleaners over residents with access to the local cleaner only. This decision rule increases the cleaners' food intake as in line with biological markets theory [25], the former would use their partner choice options and switch to another cleaner if ignored, while the latter have to wait [26]. The task of choosing between visitors and residents can be replicated in the laboratory using two Plexiglas plates with equal amounts of food, distinguishable by colour and pattern differences, are offered simultaneously, where one (the 'visitor') is removed if not inspected first, while the other plate (the 'resident') remains until the cleaners has eaten the food [24]. Cleaners excel in this task, outperforming various primate species, rats and pigeons [27–29] and being on a level with African grey parrots [30]. The task can hence be seen as a useful test for cleaners' strategic sophistication, where sensory data are integrated with decision-making algorithms that produce a specific behavioural output [31]. Therefore, we asked how far current CO₂ levels already impair cleaners' cognitive performance compared to pre-industrial levels, and to what extent predicted increase of CO₂ levels can put their performance in jeopardy. Experiments started after 30 days of acclimation to five different pCO₂ levels: (1) 275 µatm (pre-industrial mean); (2) 315 µatm (1959 mean); (3) 404 µatm (2016 mean); (4) 750 µatm (RCP 8.5 2080) and (5) 980 µatm (RCP 8.5 2100) [14].

2. Material and methods

(a) Acclimation conditions

We used cleaner wrasses *Labroides dimidiatus* ($n = 60$; size 5.5 ± 0.4 cm) collected in the Maldives and transported by TMC to the facilities of Laboratório Marítimo da Guia (Portugal). Each fish was exposed for 30 days to one of the five following pCO₂ treatments (in separated individual tanks; 12 *L. dimidiatus* per CO₂ level, three fish per mixing system): (1) 275 µatm; (2) 315 µatm; (3) 404 µatm, (4) 750 µatm; and (5) 980 µatm [17] (electronic supplementary material, table S1). A detailed description of the acclimation conditions is available in the electronic supplementary material.

(b) Learning experiment

Cleaners were tested for their ability to solve a 'biological market' task, i.e. they had to learn to prioritize feeding on an 'ephemeral' over a 'permanent' plate (see [24,27]) representing client categories based on their access to cleaners. Visitor clients have access to multiple cleaning stations were represented by the 'ephemeral' plate, and residents have access to a single station, being represented by the 'permanent' plate. In nature, visitors have been observed to leave and visit another cleaner if made to wait [26]. By contrast, residents do not have this option and hence must wait for inspection. Therefore, when these two clients arrive to a cleaning station simultaneously, cleaners have been observed to prioritize visitor clients over residents [24]. Similarly, the ephemeral plate would be retrieved if not inspected first, while the permanent plate would remain accessible until the cleaner ate the food off it. As both plates offered the same amount of food (one mysid shrimp), cleaners doubled their food intake if they inspected the visitor plate first.

Plates (8 cm × 4 cm) differed in orientation and colour of decorative stripes (i.e. horizontal blue and vertical green stripes). In each trial, both plates were presented simultaneously to a cleaner by an experimenter who was blind to the treatment (CO₂ levels). Each plate's role was pre-defined, and plate positions (i.e. left or right) were counterbalanced over 10 successive trials. The cleaners' choice was scored as 'correct' if it inspected the visitor plate first, and 'wrong' if it inspected the resident plate first. Trials continued until a cleaner reached the task criterion, defined as showing a significant prioritization over the 'ephemeral' plate, at least nine out of 10 trials, two consecutive eight out of 10, or three consecutive seven out of 10 correct choices. Following Salwiczek *et al.* [27], individuals that could not solve the task within a total of 100 trials were categorized as 'failed'. The trials were conducted over 5 days. For each trial, cleaners were first confined in a 'waiting' section of the aquarium with an opaque partition, and then, after the plates were placed on the opposite end, the partition was lifted and the cleaners had to make a choice. Past experiments have shown that the fastest cleaners to reach task criterion after 30 trials are also able to reach it again task criterion when the roles of the two plates are reversed [24,27], while those that reach it before 30 trials are not. Thus, reaching criterion between 30 and 100 trials is indicative of learning. By contrast, reaching task criterion in 10–20 trials is indicative of a colour/pattern preference for the ephemeral plate. As we are not interested in the latter, we present our results once with all individuals and once with the 'high performers' removed.

(c) Statistics

Data exploration was performed according to Zuur *et al.* [32]. The binomial data (success/failure in solving the task within 100 trials) were analysed with a Bayesian generalized linear model (BayesGLM) using pCO₂ and plate colour as covariates and a binomial distribution, as the data did fit the binomial model assumption [33]. Significance of covariates was tested

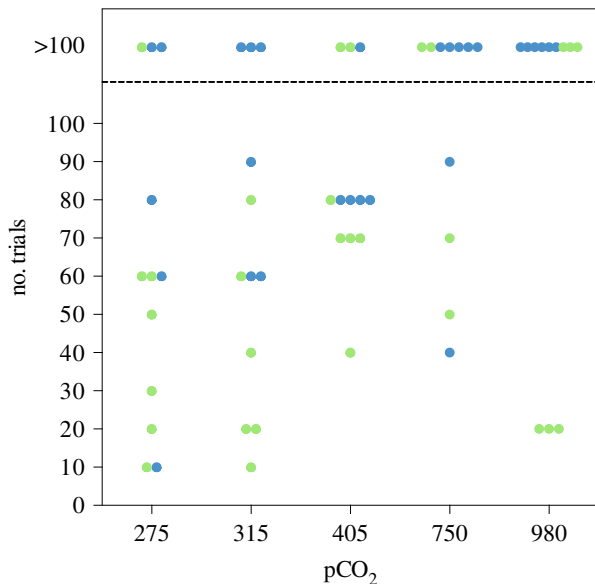


Figure 1. Cleaner wrasse cognitive performance over CO₂ concentration. The number of trials that each fish (individual dots) needed to complete the task per pCO₂ scenario; dots above the dashed line represent fish that failed to complete the task in 100 trials; colour in dots represent the colour of the correct plate.

using ANOVA. Model assumptions, namely independence and absence of residual patterns, were verified by plotting residuals against fitted values and model dispersion. Treatment level effects were verified plotting marginal effects from model terms. Statistical analysis was performed in R [34] using the HighstatLibV10 library (Highland Statistics) [35].

3. Results

Overall, we found a significant effect of CO₂ treatment ($\chi^2 = 11.355$, d.f. = 4, $p = 0.023$, figure 1) but also of plate colour on the cleaners' performance in the task ($\chi^2 = 5.629$, d.f. = 1, $p = 0.018$). The green plate as visitor yielded a higher performance than the blue ($z = 2.261$, $r^2 = 0.268$, $p = 0.023$). Most importantly, eight out of nine individuals that reached criteria within 10–20 trials were exposed to the green plate as the visitor. As all these individuals scored at least seven correct choices during the first 10 trials, we consider it more parsimonious that colour preferences rather than learning of the task is the cause for reaching criterion. If we remove the nine individuals from the analysis, the overall significant effect of treatment remains ($\chi^2 = 16.61$; d.f. = 4; $p = 0.002$) and the effect of the plate colour is lost ($\chi^2 = 1.08$; d.f. = 1; $p = 0.299$), giving us confidence in the overall effect of CO₂ manipulation on cleaner performance. Analysing the results in more detail, there is no indication that pre-industrial (275 μatm) or 1959 (315 μatm) average CO₂ concentrations would yield a higher cognitive performance than current conditions ($z = 0.248$, $r^2 = 0.266$, $p = 0.804$; electronic supplementary material, table S2). Thus, over evolutionary timescales cleaners could cope with the changes so far. The overall significant effect of CO₂ variation is driven by low performance of cleaners under high CO₂ concentrations, in particular in the 980 μatm condition ($z = -2.289$, $r^2 = 0.266$, $p = 0.022$, electronic supplementary material, table S2). In that condition, there was not a single individual that reached criterion in a way that we can exclude colour preferences being the cause. In contrast, four out of 12

individuals performed well, i.e. learned to solve the task, under 750 μatm pCO₂, without a significantly lower performance than current conditions ($z = -1.455$, $r^2 = 0.266$, $p = 0.1458$, electronic supplementary material, table S2).

4. Discussion

We observed that acclimation to past CO₂ levels did not yield a higher cognitive performance than current concentration, while the increase of CO₂ beyond current levels decreased cleaners performance. Our data suggest that there is variance in CO₂ tolerance, that such variance allowed cleaners to adjust to current levels so far and that as long as pCO₂ does not go beyond 750 μatm, and if selection on high cognitive abilities and associated sensory functions is strong enough, tolerance could spread in the population. In fact behavioural tolerance to 750 μatm of CO₂ and its heritability was previously described in other species [36–39]. This tolerance may be linked to differences in gene expression related to acid–base regulation as previously described in coral reef damselfishes [40]. However, our data also indicate that adaptation to even higher CO₂ levels is less likely as we observed no tolerance under the highest CO₂ levels, thus adaptation would likely require adaptive mutations. Cleaners' low performance in the biological market task has been documented before, namely when they occur at low density, either naturally or after environmental perturbations [2,24]. Under such circumstances, visitor clients rarely swim off if made to wait, which means that cleaners should stop caring about giving them priority of access [2]. In our experiment, all cleaners came from the same location and generally performed well under pre-industrial and current-day CO₂ concentrations. Therefore, the observed loss of cognitive performance was directly linked to a further increase in CO₂ concentrations.

Ocean acidification also impaired damselfishes' ability to learn to identify predators, an effect that was reversed with gabazine, a GABA_A receptor antagonist [9]. Those results support the hypothesis that CO₂ could alter fish cognitive performance by disruption of GABAergic neurotransmission [10]. Nevertheless, high CO₂ impairments on cleaner wrasse cooperative behaviour were recently found to be correlated with dopamine and serotonin, thus other neurotransmitter systems should also be considered as potential mechanisms of CO₂ impairments [41].

In conclusion, travelling back in time adds an extra dimension to research how climate change affects animal behaviour and underlying mechanisms. Our study suggests that the effects of increased CO₂ on fish cognition and associated sensory functions have the potential to be diminished through adaptation if CO₂ is kept under 750 μatm. This finding gives confidence that, as long as humankind is able to successfully meet the goals of the Paris Agreement (i.e. lower greenhouse gases emissions so that the world does not experience a warming higher than 1.5°C), cleaner wrasse cognition should not be affected by acidification. Indeed, under this scenario, atmospheric CO₂ concentration should not go beyond 448 μatm [42] while some cleaners could cope with CO₂ concentration of 750 μatm. Nonetheless, the interaction effects of other climate change-related stressors (such as temperature) in fish cognitive performance has not been tested, and the exposure to multiple-stressors could reduce the probability of adaptation through directional selection.

Ethics. Research was conducted under approval of FCUL animal welfare body (ORBEA) and DGAV (permit 2018-05-23-010275) in accordance with the requirements imposed by the Directive 2010/63/EU on the protection of animals used for scientific purposes.

Data accessibility. Data are available from Figshare (doi:10.6084/m9.figshare.9101993) [43].

Authors' contributions. J.R.P., R.B. and R.R. designed the study and wrote the manuscript. J.R.P., R.B. and R.R. analysed the data. J.R.P., T.R. and R.R. designed CO₂ control system. J.R.P., M.B. and F.C. performed the experiment and collected the data. All authors

discussed the results and commented on the manuscript. All authors agree to be held accountable for the content therein and approve the final version of the manuscript.

Competing interests. We declare we have no competing interests.

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